Deployment Precision Calibration

Frozen HYB-MS +10% ensemble • identity-disjoint calibration only

GATE FAILED

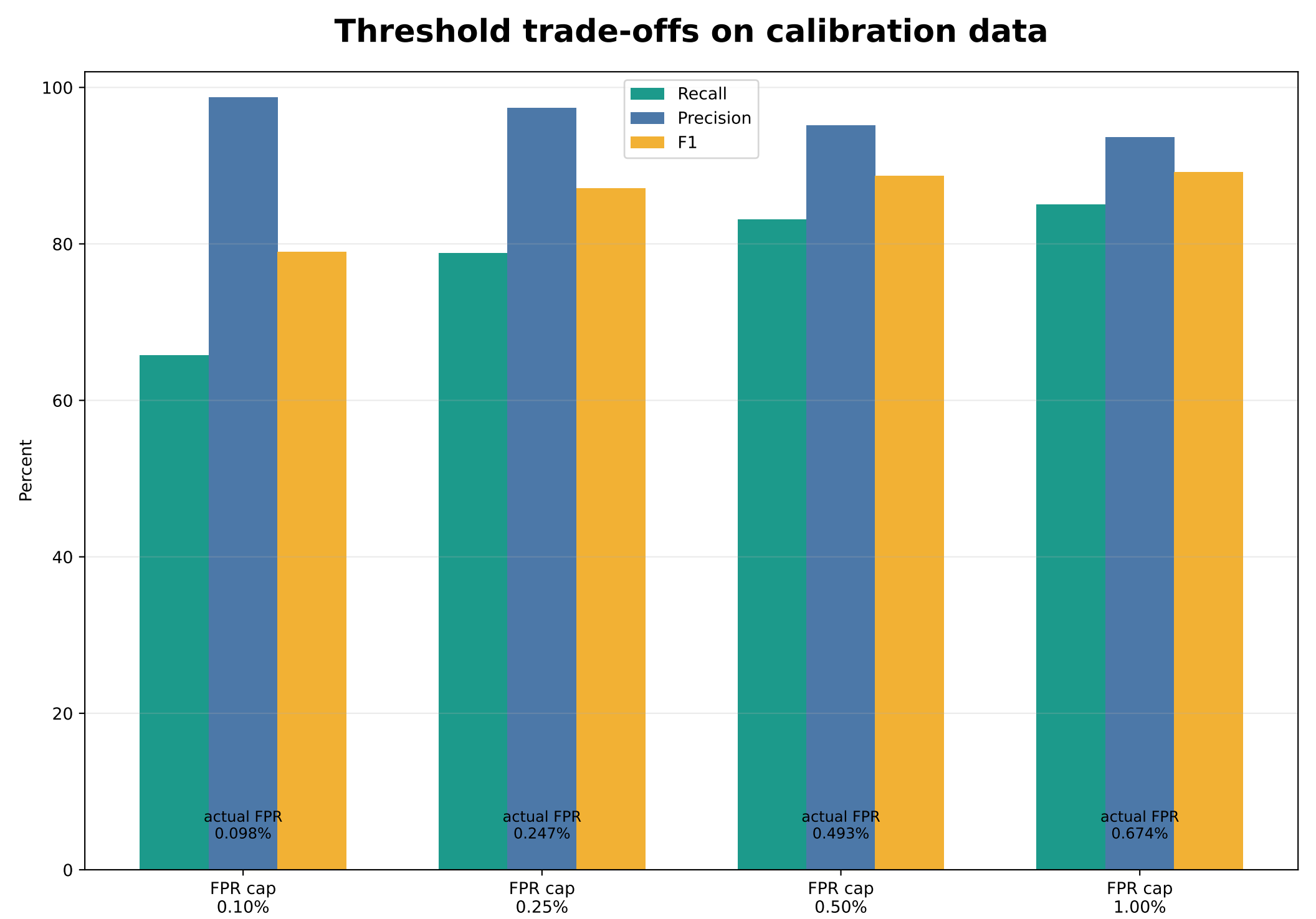
Selected threshold: 0.996419

Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
97.57%	97.38%	78.80%	87.11%	97.01%	92.83%	0.25%	21.20%

Gate: FPR ≤ 0.25% and recall ≥ 80%. Observed: FPR 0.25%, recall 78.80%.

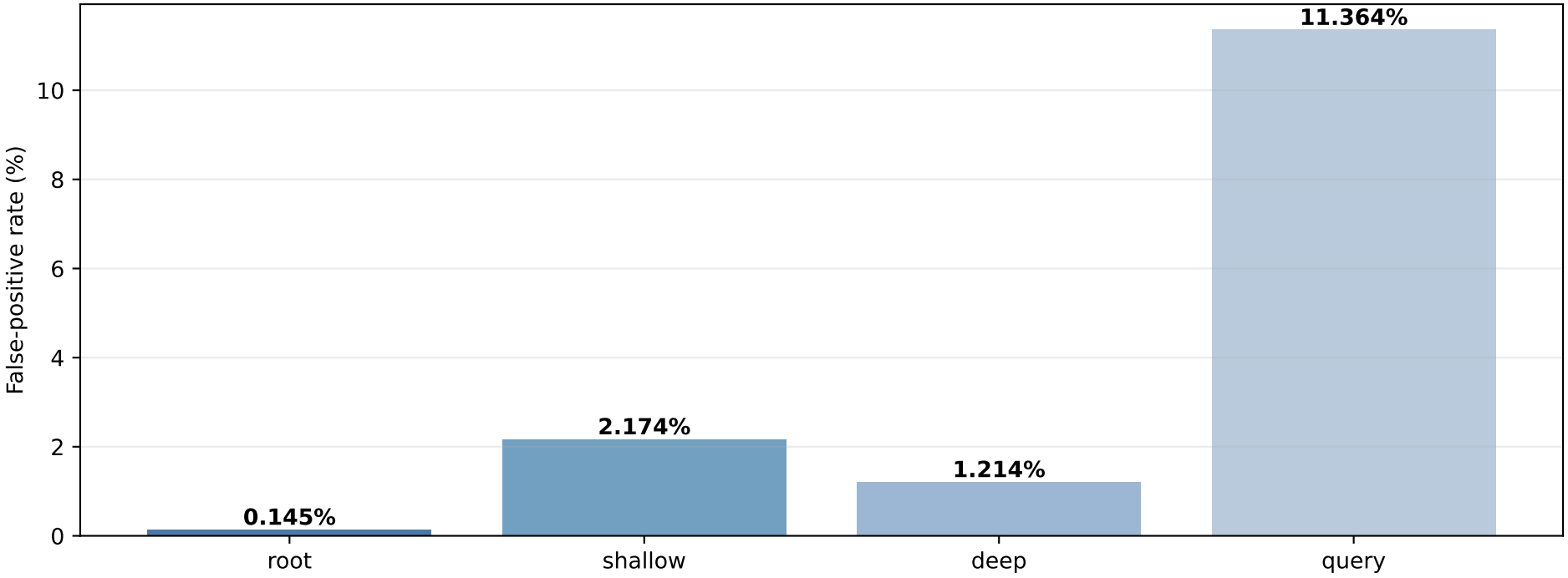
Counts: TN 21,447 • FP 53 • FN 530 • TP 1,970

This threshold was chosen without using the final confirmation set. If the gate fails, the model must be retrained using new development data and recalibrated on a separate cohort.



Original threshold 0.989095: recall 84.00%, FPR 0.60%. Selected deployment threshold is 0.996419.

Where False Positives Occur



Feature-pattern counts among false positives

none of registered flags: 21 • punycode: 9 • deep path: 16 • security words: 2 • query string: 5 • many subdomains: 1

These categories identify likely collection or representation weaknesses; they are diagnostic evidence, not permission to tune against the untouched confirmation set.

Operational Interpretation

At 0.1% phishing prevalence	**projected precision 24.24%**
At 1% phishing prevalence	**projected precision 76.35%**
At 5% phishing prevalence	**projected precision 94.39%**
At 10% phishing prevalence	**projected precision 97.26%**

Retraining decision

Gate failed; threshold adjustment alone is insufficient. Retrain with newly collected, structurally diverse legitimate development rows, then recalibrate on a separate unused cohort.

Next locked action

Apply this exact ensemble and threshold once to a newly frozen, identity-disjoint low-prevalence confirmation cohort. Do not revise the threshold after viewing those labels.